

Data management strategy in the cloud

The initial step to creating a data management strategy in the cloud is to understand the requirements of an enterprise well. Every company requires different steps to implement an effective strategy. However, some steps stay constant. Here is a list of some essential strategies.

Understand your goals: It is crucial to build data personas to clarify your company's objectives. Data personas represent detailed segments of customers/users in your target audience. Creating these personas helps the workforce to know your consumers more personally, which helps improve conversion rates. You can consider the following data points while making personas for extensive quantitative research:

- Demographics
- Purchasing behaviours
- Affiliations
- Preferences
- Buying patterns

This way, data is managed effectively to produce the most productive results.

Now, as evident, different departments will have different data needs. It should be clear as to who can access them and who cannot! So you must have specific policies for all personas accordingly.

Implement data governance:

Complying with the data governance policies is a must. All organisations and enterprises must include these policies in the data management strategies in the cloud. As a matter of fact, most companies choose data management on the cloud due to the impeccable safety provisions. Hence, these rules ensure the safety of the data as well. Some of the common compliance frameworks are listed below:

- HIPAA
- GLBA
- PCI-DSS
- FINRA
- SOX, and more

Five mistakes to avoid with cloud data management

1. Not making an apt strategy for data protection
2. Not focusing on monitoring cloud performance
3. Not focusing on the risk of cloud sprawling
4. Avoiding role-based access control
5. Ignoring a high-speed internet connection

Understand which cloud suits you best: There are different clouds, such as public, private, hybrid, and community clouds, each of which has varied utilisation. It's crucial to understand which one of these clouds will suit your requirements best.

- **Public cloud:** Third parties manage these types of clouds. Public clouds provide reliability, flexibility, high scalability, disaster recovery, and cost reduction. Additionally, public cloud services are a solution for minimal IT infrastructure costs. This option is best for small enterprises that do not have significant investments.
- **Private cloud:** These clouds work on private infrastructure and offer enterprises a certain amount of control over the resources. Private clouds provide high security, compliance with standard procedures, and more. There are various top private cloud providers, such as HP data centres, Microsoft, Ubuntu, etc.
- **Hybrid cloud:** Hybrid clouds or heterogeneous clouds combine the features of public and private clouds. They offer cheap cost, fast speed, and guaranteed safety.
- **Community cloud:** These clouds are a combination of different clouds and tackle the needs of different organisations (industry, business, community, etc). All these organisations share the same infrastructure in a community cloud. Additionally, they are scalable, adaptable, secure, and cost-effective. Generally, the healthcare industry, energy industry, media industry, and scientific organisations use this cloud.

Maintain authentic data and focus on backups: Always ensure keeping valid and authentic data. Along with that, it is necessary to track regular backups. If your company hosts its own cloud, make it mandatory for the department to run checks and make a note of the status of backups. Most cloud providers offer automated backups.

Focus on cloud data management interface (CDMI): The cloud data management interface, or CDMI, is an international standard that helps provide an interface to manage and access cloud storage. This interface comprises a comprehensive object storage model. Further, the CDMI standard has various security mechanisms to protect data. It uses the Transport Layer Security protocol to provide safety between the client and the CDMI server.

Wrapping up

You must keep your objectives clear while forming a data management strategy in the cloud. Remember that the process of managing data is ever-evolving, and you must monitor it at regular intervals.

Although there is no rule book to manage cloud data, there are some essential steps that you must take for an effective data management strategy in the cloud. I hope the steps mentioned above help you form a data management strategy that helps you in the best possible way! **END** 🐧

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